

Sydney Housing Price Prediction and Decision Support System

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SIT307 Machine Learning, 8.1 Distinction Task

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1 Part 1: Problem Definition and Data Collection

A real estate agency needs a tool that estimates sale price from listed attributes and is honest about the confidence each deserves. Three suburbs span different markets: Mosman, premium and harbourside; Parramatta, a middle ring apartment centre; and Liverpool, an affordable outer corridor, distinct in Table 1.

Listings were collected by hand from the sold sections of realestate.com.au and domain.com.au. No crawler was run, since both sites prohibit automated scraping and the task asks for manual construction. Every row stores its listing URL and collection date, so any figure here is traceable, and a blank means the listing published nothing, never an estimate.

Cleaning mattered: fifteen properties appeared on both sites under different URLs and would each have carried double weight across validation folds.

Table 1: Collection and cleaning, and the resulting suburb markets

Stage	Rows	Suburb	Properties	Median price
Collected from both sites	250	Mosman	78	\$1,251,250
Duplicated across sites	−15	Parramatta	70	\$600,000
Non standard dwelling type	−4	Liverpool	80	\$538,500
Stored dataset	231			
Removed in cleaning	−3			
Used for modelling	228			

1.1 Quality, bias and limitations

- **Apartments dominate**, 195 of 228 rows against 21 houses, with no Parramatta houses, leaving house pricing poorly learnt.
- **No description text**, forfeiting the optional text data extension.
- **Area is sparsely reported** and, as Part 2 shows, meant two different things.
- **One market period**, all sales within 206 days.

2 Part 2: Data Understanding and Feature Engineering

Prices are heavily right skewed, at 4.6, driven by Mosman sales reaching \$23,000,000. Logs reduce this to 0.6 (Figure 1), the most consequential decision in Part 3.

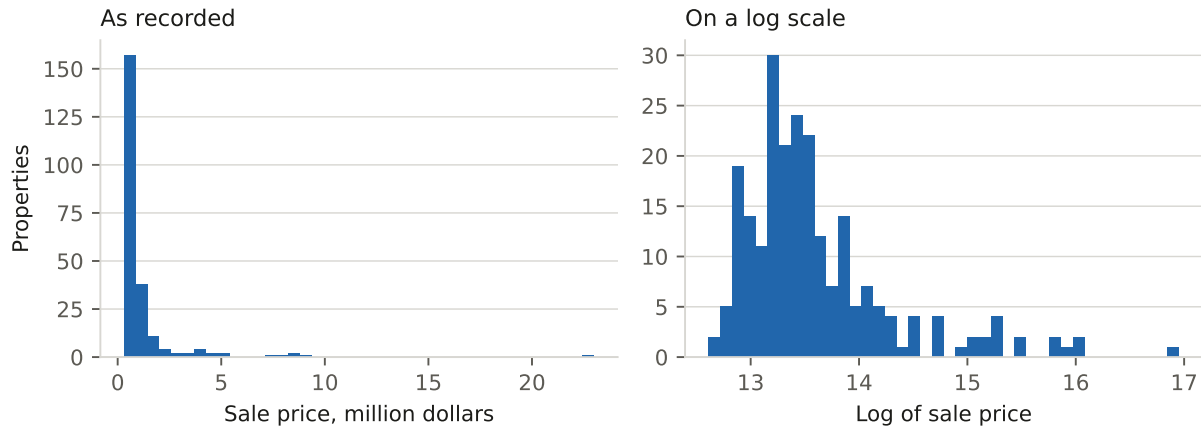


Figure 1: Sale price as recorded and on a log scale. The long right tail is what makes the raw target difficult to model.



Figure 2: Three separated markets, and attributes that each explain far less than the suburb does.

2.1 Two findings that changed the feature set

Both sites report an area figure, but it is land for a house and floor area for an apartment, and both arrived in one column. Left together the model would read the gap between roughly 600 and 100 square metres as a land effect when it is really house against flat, so it was split by property type.

Sale date looked useful, correlating at -0.40 , apparently a sharp market fall. Within each suburb that collapses towards zero: Mosman spans the full window while the others cover only its recent third, so an early date stood in for Mosman, the expensive suburb. The trend describes collection, not the market, and was excluded.

2.2 Expected drivers and engineered features

The three attributes expected to matter most were suburb, whose medians differ fourfold; property type; and bedrooms. Figure 2b supports this, every attribute correlating far more weakly than the suburb gaps imply. Distance to the city was excluded as it duplicates the suburb label exactly. Engineered features are modest: the area split, and bathrooms per bedroom.

3 Part 3: Model Development and Evaluation

Three approaches were compared: ridge linear regression as an interpretable baseline, a random forest for interactions between suburb, type and size, and gradient boosting as usually the strongest on structured data. Both trees were expected to beat the baseline clearly.

The target was tested first. Predicting log price improves every model and rescues gradient boosting from an R^2 of 0.047 to 0.544, because on raw dollars the squared error is dominated by a few multi million dollar sales. All results use five fold cross validation on that target, scored back in dollars.

Table 2: Five fold cross validation. The two R^2 columns describe the same predictions, measured in log space and in dollars.

Model	Log R^2	Train R^2	Gap	Dollar R^2	Median error
Linear Regression	0.869	0.906	0.038	0.402	14.5%
Random Forest	0.872	0.961	0.089	0.537	11.4%
Gradient Boosting	0.876	0.962	0.086	0.544	11.7%

The expectation was only partly correct. On the log scale all three are almost indistinguishable, linear regression falling within 0.01 of both trees while showing the smallest train and test gap. With seven features and no text, little non linear structure remains, because logs already convert multiplicative price behaviour into something additive. The trees separate only on the dollar metrics, which weight expensive errors heavily. Random forest is recommended: its 11.4% median error is lowest and is the metric the tool reports. Quoting only the log R^2 of 0.87 would overstate it; 0.54 is what matters to somebody handed a price.

4 Part 4: Investigating Prediction Failures

Table 3: The five largest errors, from cross validated predictions

Property	Type	Beds	Sold	Predicted	Error
17 Morella Road	House	4	\$23,000,000	\$5,200,000	77%
34 Rickard Avenue	House	7	\$9,000,000	\$4,700,000	48%
96 Glover Street	House	4	\$3,390,000	\$7,300,000	116%
5 Botanic Road	House	4	\$8,700,000	\$5,300,000	39%
G05 Myahgah Road	Apartment	2	\$4,500,000	\$1,700,000	62%

All five are in Mosman and the errors run both ways: three trophy houses under predicted, 96 Glover Street over predicted by more than double, and the fifth sold for four times the Mosman apartment median, almost certainly a penthouse but recorded as a two bedroom flat.

The common thread is not that these are expensive but that in Mosman the recorded attributes stop discriminating. Only 13 Mosman houses span \$3,390,000 to \$23,000,000 and six publish no land size, so the model separates them on bedroom counts that barely differ. Position, outlook and build quality set these prices, and none are in the data.

Figure 3 quantifies it: 21.2% median error in Mosman against 8.2% and 9.7% elsewhere, not because Mosman is costly but because its stock is heterogeneous with few comparable sales. The most valuable missing information is recent comparable street sales, what a valuer consults first.

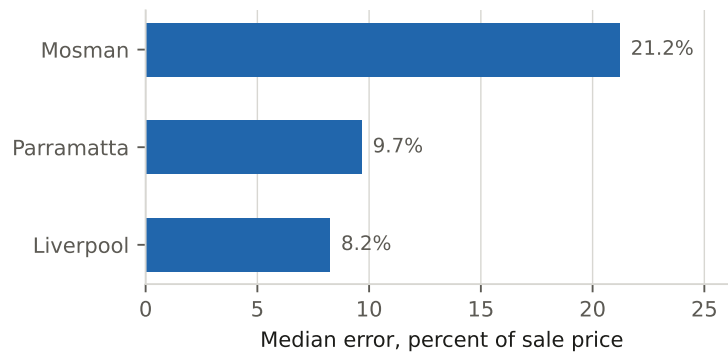


Figure 3: Median percentage error by suburb, from cross validated predictions. Mosman is roughly two and a half times harder.

5 Part 5: Final Deployment and Reflection

5.1 How the application was built

The tool is a Flask application, keeping prediction logic in one small Python file while the page stays plain HTML and CSS, so it runs with no build step. On start it loads the saved random forest and a schema of the feature list, per suburb error rates and thin combinations.

Three decisions shape it. Inputs are validated on the server, where untrusted data enters. The result is a range, set by that suburb's cross validated median error. And it refuses to bluff: asked for a Parramatta house it says so (Figure 5).

SP

Sydney Property Price Estimator

228 recorded sales

Estimate a sale price

Enter the details of a property in Mosman, Parramatta or Liverpool. Estimates come from a model trained on 228 real sales collected from realestate.com.au and domain.com.au, and are shown with the range the model is typically accurate to in that suburb.

Location and type

Suburb

Liverpool

Property type

Apartment

Sale method

Private treaty

The property

Bedrooms

2

Bathrooms

1

Parking spaces

1

Floor area

Square metres of internal floor area. Optional, often not listed.

Estimate price

Estimated sale price

\$410,881

Likely range \$377,189 to \$444,573

In Liverpool this model is typically within 8.2 percent of the sale price, measured by cross validation on sales it had not seen.

This is a guide, not a valuation. The model cannot see position, outlook, renovation standard or recent comparable sales, which is what separates two otherwise identical homes.

Figure 4: The application estimating a Liverpool apartment. The range comes from the 8.2% median error measured for that suburb.


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Location and type

Suburb

Parramatta

Property type

House

Sale method

Private treaty

The property

Bedrooms

4

Bathrooms

2

Parking spaces

2

Land size

500

Square metres of land.
Optional, often not listed.

Estimate price

Estimated sale price

\$1,019,303

Likely range \$920,431 to \$1,118,176

There are no house sales in Parramatta in the training data, so this figure is an extrapolation from other suburbs and types. Treat it as a rough indication only.

In Parramatta this model is typically within 9.7 percent of the sale price, measured by cross validation on sales it had not seen.

This is a guide, not a valuation. The model cannot see position, outlook, renovation standard or recent comparable sales, which is what separates two otherwise identical homes.

Figure 5: Asked for a Parramatta house, a combination absent from the training data, the tool warns instead of answering confidently.

5.2 Building and running it

```

pip install -r requirements.txt

# Rebuild the dataset, features and notebook
python3 scripts/prepare_features.py
jupyter nbconvert --to notebook --execute --inplace \
    notebooks/sydney_housing_price_prediction.ipynb

# Run the web application, then open http://127.0.0.1:5000
python3 app/flask_app.py

```

Select a suburb and type, enter the attributes and press Estimate price. Choosing a house switches the area field from floor area to land size.

5.3 Reflection

The lessons came from the data, not the algorithms. The findings above surfaced only because results were checked rather than accepted, none were visible in summary statistics, and each would have quietly damaged the model. Choosing the log target mattered more than the algorithm.

Accuracy is unevenly distributed, dependable for Liverpool and Parramatta apartments and unreliable for Mosman houses. Presenting both with equal confidence would mislead a buyer precisely where the sums are largest, which is why the tool reports suburb ranges and flags thin combinations. The dataset is also 85% strata.

With more time the priorities would be collecting the descriptions and build years listing pages publish, balancing property types, and adding comparable street sales.

Acknowledgement of GenAI use

AI assistance was used to plan, write and review code across the notebook, the collection tooling and the web application, and to improve the quality of the written text and code. Every number, table and figure comes from executing that code on the collected data; none were estimated. The analytical decisions are explained above.

Code and data

All code, data, the notebook and application are at:

`https://github.com/Shalok-sys/Sydney-Housing-Price-Prediction`

It contains `data/` (collected listings and cleaned table), `notebooks/` (full analysis with visible outputs), `scripts/` (collection, validation, feature pipeline), `app/` (the Flask application) and `models/`.

References

- [1] Pedregosa, F. et al. (2011). Scikit learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825 to 2830.
- [2] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), pp. 5 to 32.
- [3] realestate.com.au (2026). Sold property listings, Mosman, Parramatta and Liverpool NSW. Accessed September 2026.
- [4] domain.com.au (2026). Sold property listings, Mosman, Parramatta and Liverpool NSW. Accessed September 2026.